PROJECT REPORT ON

PREGNANCY DATA

PREPARED BY

1) D.G.SANDANI GUNASEKARA -S16903

2) H.S.S PERERA - S16661

3) W.A.D.FERNANDO - S16979

4) J.B.A.A.Y DHARMASENA - S16938

5) O.S. JAYATHUNGA - S16807

Introduction

The pregnancy dataset is concerned with birth weights. The data set is derived from pregnancies that occurred in the San Francisco East Bay area between 1960 and 1967. It is very important to figure out what the most important factors influencing birth weight are to develop child health for governments and any other health service providers.

Our objective is to investigate how the predictor variables (gestation, parity, age, height, weight, and smoking) will influence the interest variable (Birth Weight). Using this kind of analysis, we can build up a model to predict Birth Weights. Descriptive data analysis and regression analysis will identify how predictor variables influence the birth weight.

The data set that we selected has 8 predictor variables and 1236 observations. We are willing to model available data in many ways to get a much better picture about our interest variable. We are interested to see the model of predicting Birth Weights and classifications of birth weights, classified categories are, Underweight (less than 88 ounces) , Normal weight (between 88 – 141 ounces) and Overweight (greater than 141 ounces).

Objectives of the project

1.Find out influential factors on Birth Weight.

2.Examine what are the most important factors to keep birth weight in the healthy range (between 88 and 141 ounces)

3.Model the data in a way that R^2 (coefficient of determination) is as high as possible without over fitting.

Source - https://www.kaggle.com/datasets/debjeetdas/babies-birth-weight

Literature Review

1.Birth weight prediction : By strawberries4life

Overview -

This notebook includes EDA and randomforest regressor model.The author examines the distribution of birthweight(bwt) and any impact that such demographic factors as gestation,parity,age,height,weight and smoke may have on birthweight .

Key observations-

- There is a positive correlation between height and weight, one possible new feature could be BMI
- There seems to be a data imbalance due to the parity feature, but this feature does not have much correlation with birthweight
- Babies born to mothers who smoke are more likely to have low birthweight. Also the data has quite a few outliers.

2.PregnancyDataAnalysis : By Abdullah shahzad

Overview -

Initial analysis including count , mean , standard deviations , percentiles for variables birthweight, gestation , age , height and weight has been carried out in this notebook. Correlations between above variables and scatter plots are also included.

Key observations-

- A high gestation period leads to high birthweight
- There is no significant difference in the traits of smoker and non smoker womens
- BWT Not depend on the age, height or weight
- Age, weigth and height are as usual normaly distributed(Natural)

3. smokingVSpregnancy : By Stepan Dolzhenko

Overview -

Data analysis, statistical analysis and bootstrapping was performed to analyses the pregnancy dataset. The missing values of the height and weight of the mother have been replaced by the averages when doing this analysis. Gestation age is considered as connected to the body weight of infants, so it is excluded from thesis of this research.

Key observations-

- There are both premature, preterm & low weighted children and premature macrosomic children.
- When analyzing data, it was found out that children who's mothers smoke have higher chance to be born preterm.
- Smoking can be connected with the birth weight, so this feature is more relevant than gestation.

Adequacy of Findings

Adequacy	Analysis of this report	Other Kaggle Articles
1. Model Adequacy	• Conducted forward selection in multiple regression model based on their statistical significance. • Checked for multicollinearity to ensure that the model doesn't include high correlated predictors.	• Strawberries4life: Random forest regression modelling has been implemented and the model has been evaluated using RMSE metrics for better accuracy. • Abdullah shahzad: Mainly focused on exploratory data analysis. Modelling has not been conducted.
2. Residual Analysis	• Conducted residual analysis to check if any assumptions have been violated. • The residual vs fitted values plot does not show any pattern which indicates linearity and constant variance of residuals. • The Q-Q plot is strongly linear which indicates normality of residuals.	• None of the kaggle articles relating to this dataset had residual analysis.

Descriptive Analysis

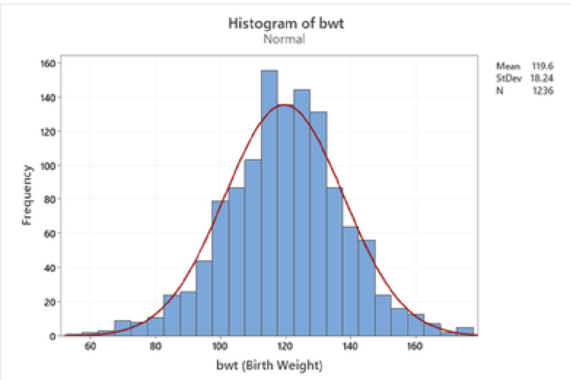
The “Pregnancy dataset” containing 8 variables describe the infant birthweight. variables include a unique id (case) and other factors that can affect the birthweight such as gestation, parity, age, height, weight and smoke. This dataset allows analysis to conduct deep analyses and model which factors influences the birthweight of the infant.

1. **case** - Each individual has a unique ID number
2. **bwt** - Birthweight of the baby in Ounces range from 55 to 176
3. **gestation** - Length of gestation, in days
4. **parity** - Binary indicator for a first pregnancy(0 = first pregnancy, 1- not the first pregnancy)
5. **age** - Mother’s age in years ranged from 15 to 45
6. **height** - Mother’s height in inches
7. **weight** - Mother’s weight in pounds
8. **smoke** - Binary indicator for whether the mother smokes(0-No, 1- Yes)
9. **weight-status** - Categories of bwt (Under weight - less than 88 (Ounce), Normal weight - between 88 - 141 (Ounce), Over weight - greater than 141 (Ounce)

- The graphs below show the descriptive analysis of the pregnancy dataset.

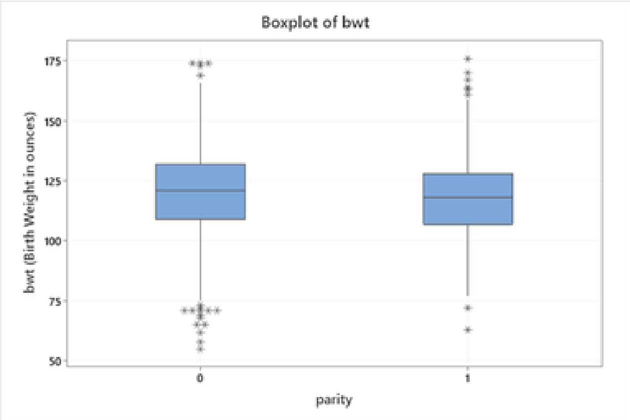
Pregnancy Factors

Histogram for Birth Weights (ounces)



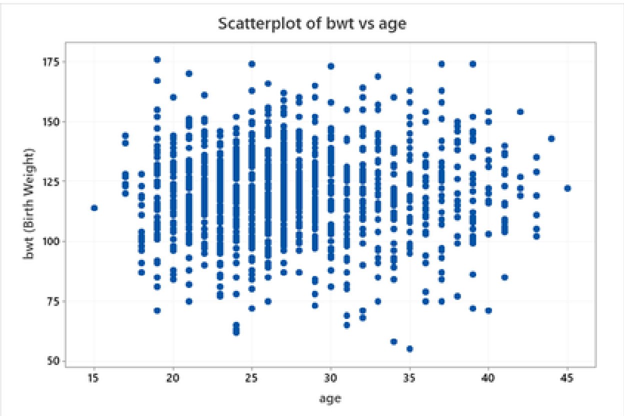
The above histogram shows a distribution that is approximately normal with a mean of 119.6 ounces and a standard deviation of 18.24. The majority of babies are born with a weight of 199.6 ounces. Birth weight is distributed around the range of 55 ounces to 176 ounces.

Box plots for Birth Weight categorized by parity



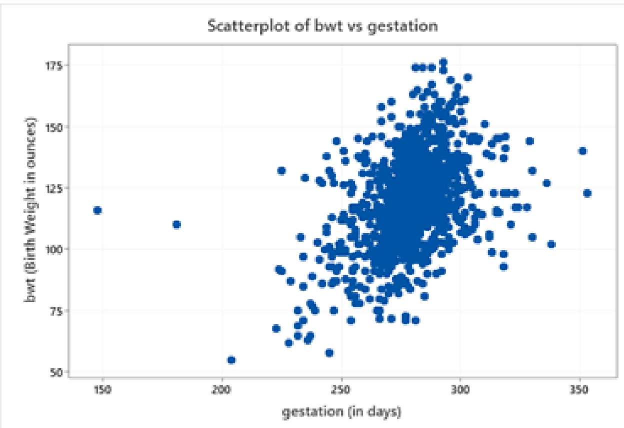
The parity 0 (0–first pregnancy) group has a slightly higher median than the parity 1 (1–not the first pregnancy) group. The parity 0 (0 – first pregnancy) group has a slightly larger interquartile range than the parity 1 (1 – not the first pregnancy) group. This suggests a bit more spread in the parity 0 (0 – first pregnancy) group. The box plot exhibits outliers on both ends of the whiskers, indicating a wider range of birth weights in the parity 0 (0—first pregnancy) group than the other group (parity 1).

Scatterplot for Birth Weight vs age of the mother



The data plots are scattered randomly, therefore, the scatterplot for the age of the mother and the birth weight of the baby does not show any correlation.

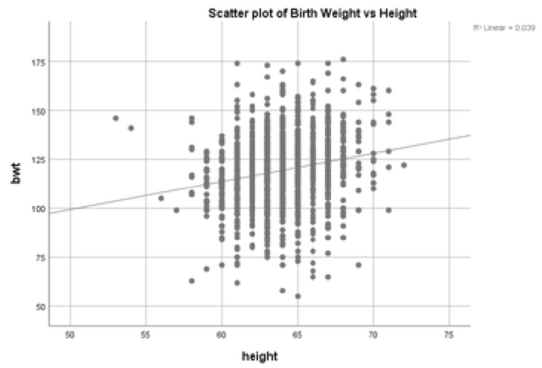
Scatterplot for Birth Weight Vs Gestation



Most of the data points are plotted around the 200–300 (in days) range for gestation, and birth weight is distributed among the range of 50-175 (in ounces). This scatterplot exhibits a low positive correlation between two quantitative variables. Which means that as the gestation period increases, birth weight also increases.

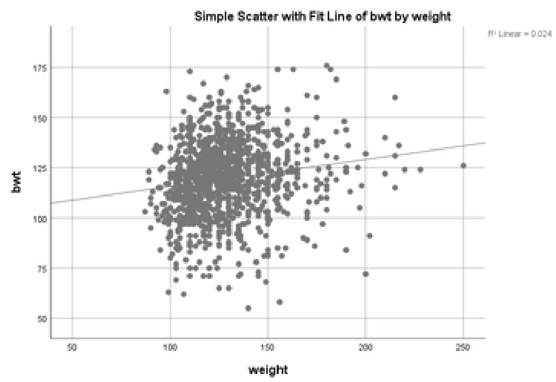
Parent health & lifestyle

Scatter plot of Birth Weight vs Height of the Mother



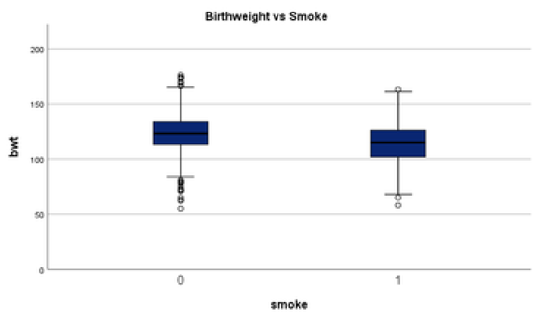
This scatter plot of Birth Weight vs Height shows a weak
positive correlation between Birth Weight and Height.

Scatter plot of Birth Weight of the Infant vs Weight of the mother



This scatter plot of Birth Weight of the Infant vs Weight of the mother shows a weak correlation between birth weight of the infant and Weight of the mother.

Box plot of Birth Weight vs Smoke



This Box plot shows the birth weight of smokers(1) and non smokers(0). There are some outliers in both smokers and non-smokers and outliers are higher in non-smokers compared to smokers. The median birthweight of babies of non-smoking mothers is slightly higher than smoking mothers and both the median birth weights are around 100-150 ounces.

Advanced Analysis

MLR Analysis

Model fitting

Forward selection was used to fit the model. According to the code below, the output was obtained.

Inserting first variable

Variable	degree of freedom	F value	P value
Gestation	1221	243.6	2e-16
Parity	1234	2.629	0.105
Age	1232	1.396	0.238
Height	1212	49.72	2.97e-12
Weight	1198	29.12	8.21e-08
Smoke	1224	74.87	2e-16

F value is larger in Gestation. Therefore it was added to the model.

New model :
bwt ~ Gestation

Inserting second variable

Variable	degree of freedom	F value	P value
Parity	1220	127.1	0.00269
Age	1218	123.1	0.0457
Height	1199	150.2	1.86e-11
Weight	1185	137.1	3.64e-08
Smoke	1210	161.9	2e-16

The Smallest P value was obtained in Smoke.

New model :
bwt ~ Gestation+Smoke

Inserting third variable

Variable	degree of freedom	F value	P value
Parity	1209	111.9	0.0021
Age	1207	108.5	0.187
Height	1188	131.9	8.59e-13
Weight	1174	116.7	2.82e-07

The Smallest P value was obtained for height.

New model :
bwt~Gestation+Smoke+ Height

Inserting fourth variable

Variable	degree of freedom	F value	P value
Parity	1187	102.5	0.000957
Age	1186	99.27	0.222
Weight	1170	98.45	0.0161

The Smallest P value was obtained for Parity.

New model : bwt~Gestation+Smoke+ Height+parity

Inserting fifth variable

Variable	degree of freedom	F value	P value
Age	1185	81.86	0.93698
Weight	1169	81.29	0.04589

The Smallest P value was obtained for Weight.

New model : bwt~Gestation+Smoke+ Height+parity+Weight

Variable	degree of freedom	F value	P value
Age	1185	67.61	0.91696

Age is insignificant because p value is greater than 0.05. Therefore it can't be added to the model.

Main Effects Model

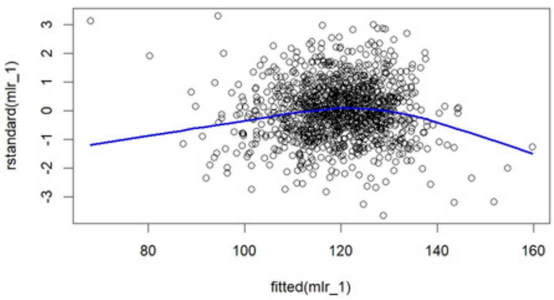
```
> summary(lm(bwt~gestation+as.factor(smoke)+height+as.factor(parity)+weight))

Call:
lm(formula = bwt ~ gestation + as.factor(smoke) + height + as.factor(parity) +
    weight)

Residuals:
    Min      1Q  Median      3Q     Max
-57.713 -10.151  -0.156   9.672  51.597

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)        -80.63568   14.03671  -5.745 1.17e-08 ***
gestation            0.44400    0.02906  15.280  < 2e-16 ***
as.factor(smoke)1   -8.38341    0.95013  -8.823  < 2e-16 ***
height               1.15378    0.20461   5.639 2.14e-08 ***
as.factor(parity)1  -3.29166    1.06231  -3.099  0.00199 **
weight               0.04999    0.02501   1.999  0.04589 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 15.82 on 1169 degrees of freedom
  (61 observations deleted due to missingness)
Multiple R-squared:  0.2579,    Adjusted R-squared:  0.2547
F-statistic: 81.26 on 5 and 1169 DF,  p-value: < 2.2e-16
```

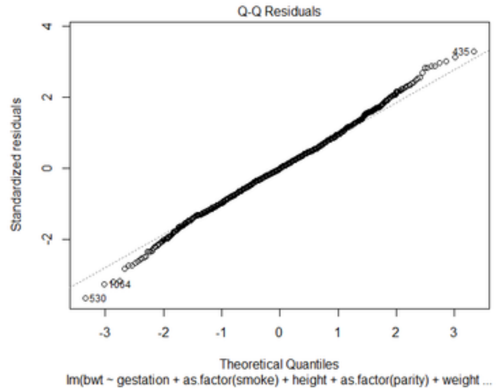
The model we obtained :

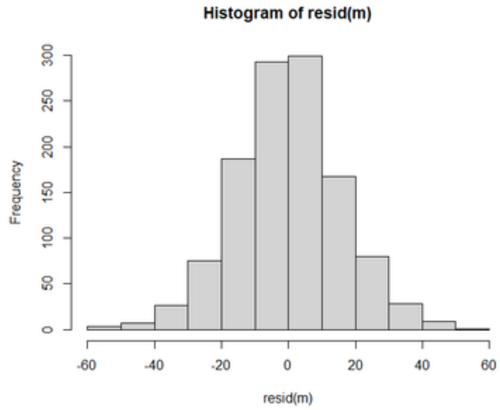
btw = (-80.636) + 0.444*gestation + (-8.383)*smoke + 1.154*height +(-3.292)*parity + 0.04999*weight

- According to the summary obtained R^2 is 25.79%. That means only 25.79% variation has been described by the model.

Checking Model Assumptions







Data points of residuals VS fitted values plot are scattered randomly around 0 line and most of the data points are gathered into mid of the graph. So this is linearly independent and has constant variance.
Blue line of this plot shows a slight curve that indicates that some
predictors like interaction terms, polynomial terms and transformations might be missing.

Q-Q plot and histogram of residuals show that the residuals are normally distributed.
Therefore transformations are not required..

Checking Multicollinearity

Multicollinearity of the main effect model

gestation	as.factor(smoke)	height	as.factor(parity)	weight
1.015045	1.010136	1.252853	1.025227	1.261596

All VIF values are close to 1 , therefore multicollinearity does not exist.

- Interaction terms were added to increase the multiple coefficient of determination, R^2.

variable	P value
gestation*parity	0.4056
gestation*smoke	0.000532
gestation*height	0.52017
gestation*weight	0.09382
parity*smoke	0.51519
parity*height	0.0578
parity*weight	0.6278
smoke* height	0.56160
smoke*weight	0.87242
height*weight	0.69054

gestation*smoke has the lowest p value. Therefore add that to the model.

new model:
bwt~Gestation +Smoke+ Height+parity+Weight+ gestation*parity

variable	P value
gestation*parity	0.378942
gestation*height	0.370310
gestation*weight	0.150596
parity*smoke	0.693902
parity*height	0.063730
parity*weight	0.611757
smoke* height	0.374594
smoke*weight	0.723468
height*weight	0.709257

All the p values are greater than 0.05 and they are insignificant. Therefore they can't be added to the model .

Model with interaction terms :

btw = (-80.636) + 0.444*gestation + (-8.383)*smoke + 1.154*height + (-3.292)*parity +0.04999*weight + 0.000532*smoke*gestation

Still R^2 is 26.17%

Linearity and independent assumptions were not violated.

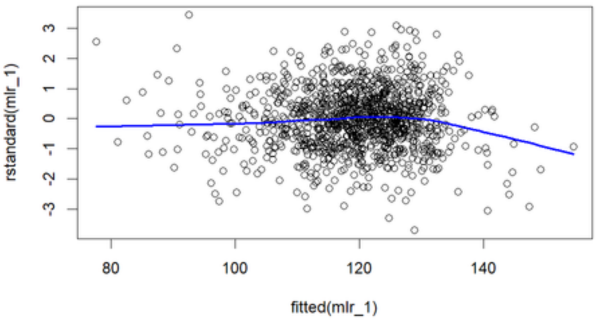
To improve the model fit, polynomial terms were introduced. Quadratic terms were tested, and the following results were obtained:

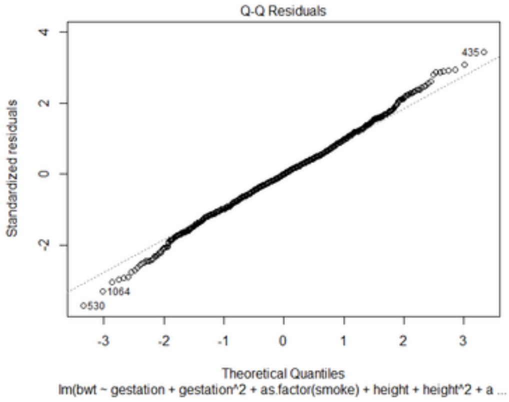
```
> mlr_2=lm(bwt~gestation+gestation^2+as.factor(smoke)+height+height^2+as.factor(parity)+weight+weight^2+gestation*as.factor(smoke))
> summary(mlr_2)

Call:
lm(formula = bwt ~ gestation + gestation^2 + as.factor(smoke) +
    height + height^2 + as.factor(parity) + weight + weight^2 +
    gestation * as.factor(smoke))

Residuals:
    Min      1Q  Median      3Q     Max
-57.872  -9.940   0.000   9.532  53.469

Coefficients:
                            Estimate Std. Error t value Pr(>|t|)
(Intercept)                -59.01455   15.29464  -3.859 0.000120 ***
gestation                    0.37022    0.03588  10.317  < 2e-16 ***
as.factor(smoke)1          -66.72407   16.82227  -3.966 7.74e-05 ***
height                       1.14747    0.20366   5.634 2.20e-08 ***
as.factor(parity)1          -3.30096    1.05732  -3.122 0.001840 **
weight                       0.04562    0.02493   1.830 0.067504 .
gestation:as.factor(smoke)1  0.20938    0.06028   3.474 0.000532 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 15.74 on 1168 degrees of freedom
  (61 observations deleted due to missingness)
Multiple R-squared:  0.2655,    Adjusted R-squared:  0.2617
F-statistic: 70.37 on 6 and 1168 DF,  p-value: < 2.2e-16
```

```
Analysis of Variance Table

Response: bwt
                              Df Sum Sq Mean Sq  F value    Pr(>F)
gestation                      1  65461   65461 264.1583 < 2.2e-16 ***
as.factor(smoke)               1  19510   19510  78.7293 < 2.2e-16 ***
height                         1  12799   12799  51.6483 1.181e-12 ***
as.factor(parity)              1   2866    2866  11.5664 0.0006942 ***
weight                         1    999     999   4.0321 0.0448727 *
gestation:as.factor(smoke)     1   2990    2990  12.0656 0.0005324 ***
Residuals                   1168 289444     248
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
>
```

- R^2 is 26.55% this means 26.55% variation of response variable is explained by this model.

Checking assumptions after adding the interaction terms





There is no any specific pattern in the standard residuals VS fitted values plot. That confirms residuals have constant variance and that the linearity assumption is not violated.
Q-Q plot indicates linear relationship that shows residuals are normally distributed.

Final Model

btw = (-59.01455) + 0.37022*gestation + (-66.72407)*smoke + 1.14747*height + (-3.30096)*parity +0.04562*weight + 0.20938*smoke*gestation

parity	dummy
first pregnancy	0
not first pregnancy	1

smoke	dummy
yes	0
no	1

used libraries

```
> library(car)
Loading required package: carData
Warning message:
package ‘car’ was built under R version 4.5.1
```

Ordinal Analysis

Weight status classification

Under weight	less than 88 (Ounce)
Normal weight	between 88 - 141 (Ounce)
Over weight	greater than 141 (Ounce)

Model fitting

Forward selection method was used to fit the model. Started with the null model and added one variable to the model at a time. Then the summary of the outputs were as below.

Inserting first variable

Variable	**p-value**
gestation	<2e-16
parity	0.242
age	0.0603
height	7.74E-07
weight	0.000823
smoke	1.79E-05

The smallest p value is obtained from gestation.

New model:

bwt ~ gestation

Inserting second variable

Variable	**p-value**
parity	0.0461
age	0.0131
height	3.90E-06
weight	0.00106
smoke	6.53E-05

The next lowest p value is of height.

New model:

bwt~gestation+height

Inserting third variable

Variable	**p-value**
parity	0.0314
age	0.011
weight	0.134637
smoke	3.00E-05

Next smallest p value is obtained from smoke.

New model:

bwt~gestation+height+smoke

Inserting fourth variable

Variable	p-value
parity	0.0382
age	0.041
weight	0.246687

Next smallest p value is obtained from parity.

New model:
bwt~ gestation+height+smoke+parity

Variable	p-value
age	0.161
weight	0.39838

Age, weight are insignificant because p values are greater than 0.05. Therefore they can't be added to the model.

The final model we obtained ;

weight_status = 0.053846 * gestation + 0.156988 * height - 0.746268 * smoke - 0.390617 * parity

- The deviance of the above model was as follows:

```
library(MASS)
fit <- polr(mlr, data = babies, method = "logistic")
deviance(fit)
```

which gives the deviance as,

```
> library(MASS)
> fit <- polr(mlr, data = babies, method = "logistic")
> deviance(fit)
[1] 1119.811
```

- Then the significance of the interaction terms were checked and the interaction term (gestation*smoke) was added. Then the deviance of the model reduced as shown below indicating a better fit of the model:-

```
fit <- polr(weight_status~gestation+height+as.factor(smoke)+as.factor(parity)+gestation*as.factor(smoke), data = babies,
deviance(fit)
```

then the deviance was,

```
> fit <- polr(weight_status~gestation+height+as.factor(smoke)+as.factor(parity)+gestation*as.factor(smoke), data = babies, met
hod = "logistic")
> deviance(fit)
[1] 1109.596
```

Final Model Output after adding interaction terms

Code:

```
summary(clm(weight_status~gestation+height+as.factor(smoke)+as.factor(parity)+gestation*as.factor(smoke),data=babies))
anova(clm(weight_status~gestation+height+as.factor(smoke)+as.factor(parity)+gestation*as.factor(smoke),data=babies))
```

Outputs:

```
> summary(clm(weight_status~gestation+height+as.factor(smoke)+as.factor(parity)+gestation*as.factor(smoke),data=babies))
formula: weight_status ~ gestation + height + as.factor(smoke) + as.factor(parity) + gestation * as.factor(smoke)
data:    babies

 link  threshold nobs logLik  AIC     niter max.grad cond.H
 logit flexible  1192 -554.80 1123.60 8(0)  1.54e-11 2.4e+08

Coefficients:
                            Estimate Std. Error z value Pr(>|z|)
gestation                   0.043026   0.006099   7.054 1.74e-12 ***
height                      0.156398   0.033462   4.674 2.96e-06 ***
as.factor(smoke)1          -8.945344   2.571485  -3.479 0.000504 ***
as.factor(parity)1         -0.389019   0.189806  -2.050 0.040408 *
gestation:as.factor(smoke)1 0.029314   0.009162   3.199 0.001377 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Threshold coefficients:
                         Estimate Std. Error z value
Underweight|Normalweight   18.151      2.724   6.663
Normalweight|Overweight    24.113      2.808   8.586
(44 observations deleted due to missingness)
```

```
> anova(clm(weight_status~gestation+height+as.factor(smoke)+as.factor(parity)+gestation*as.factor(smoke),data=babies))
Type I Analysis of Deviance Table with Wald chi-square tests

                           Df    Chisq Pr(>Chisq)
gestation                   1 116.8177  < 2.2e-16 ***
height                      1  21.1778  4.186e-06 ***
as.factor(smoke)            1  19.0335  1.284e-05 ***
as.factor(parity)           1   4.1969   0.040497 *
gestation:as.factor(smoke)  1  10.2365   0.001377 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Packages used

```
install.packages("ordinal")
library(ordinal)
```

```
library(MASS)
```

Our final model=>

log [P(y<=Underweight)/P(y>Underweight)]=18.151+0.043026*gestation+0.156398*height -8.945344*smoke-0.389019*parity+0.029314(gestation*smoke)

log[P(y<=Normalweight)/P(y>Normalweight)]=24.113+0.043026*gestation+0.156398* height-8.945344*smoke-0.389109*parity+0.029314(gestation*smoke)

Discussion and Conclusion

Discussion

To analyze the pregnancy dataset two method were used: multiple linear regression and Ordinal regression each capturing patterns and contribution from variables.

Multiple linear regression

MLR model was formed using forward selection. Variation in the birth weight of the baby is explained by gestation period, parity, smoking status of mother, weight of mother and height of mother. Although assumptions are not violated, R^2 is low. To increase R^2 , interaction terms and polynomial terms were added. Standard residuals VS fitted values plot does not give any specific pattern indicating a constant variance. Therefor, doing transformations on this model does not have much effect on R^2 . Multicollinearity of the variables is close to 1. That indicates that there is no correlation between variables. Interaction terms were added stepwise by checking significance of that interaction term through P value. To increase R^2 further, quadratic terms were added to the model.
Then R^2 value was increased to 26.55%.
There are four influential points. Removing those influential points, can obtain a model with a larger R^2 than the current value. But we couldn't able to find any justified reasons to remove those observations from our dataset. Therefore, we didn't proceed with that method.
Residual Analysis - Residual analysis was conducted to check whether the assumptions of multiple linear regression had been met. Standard residuals VS fitted values plot of final model does not indicate any specific pattern that confirming that the linearity assumption is not violated and that variance is constant. The Q-Q plot shows that residuals are normally distributed.
This helps to increase confidence in predictability and validity of MLR model.

Ordinal linear regression

An ordinal regression model was also fitted by considering the ordinal aspect of the birth weight categorization. Forward selection was used to include the suitable predictors. "Gestation" was the first variable to enter followed by "height", "smoke" and then "parity". By using an ordinal regression model where the birthweight is viewed as an ordinal categorical variable, provide a different perspective which is helpful in drawing certain conclusions to make decisions in a better way.
The deviance of the final model was found to be 1119.811. Then by checking the significance of the interaction terms using the chi squared values, the interaction term (gestation*smoke) was added. All other interaction terms turned out to be insignificant. The deviance of the model after the interaction term was added had decreased to 1109.596 which signals an improvement of the model.
This model is helpful in identifying the healthy levels of the variables for the birthweight to be in the normal birthweight range.

Conclusion

The analysis of the pregnancy dataset using both multiple linear regression and ordinal linear regression investigates the main facts of birthweight as well as the modelling challenges. The initial MLR model obtained from forward iteration method gave a very low R^2 value indicating that the included predictors in the model do not explain the variation of the birthweight. By adding interaction terms after checking multicollinearity using VIF values slightly improved the R^2 value by 1%. Adding polynomial terms didn't make a significant improvement of R^2 values and applying log transformations are not possible due to constant variance of the residuals.

To obtain a better model the response variable was categorized to underweight, overweight, and normal weight according to MEDizzy(medical learning community).The deviance obtained for the model was very high, and even after adding interaction terms, the deviance remained high. However, the ordinal model explains birthweight more effectively through weight categories.

Overall, this analysis suggests that the predictor variables included in this dataset do not have a strong direct effect on infant birth weight. Future analysis could be done by removing influential points from the model and by adding additional, more influential variables to improve model effectiveness.

Contributions

1) D.G.Sandani Gunasekara (s16903) - MLR analysis, Discussion and Adequacy of findings
2) H.S.S Perera (s16661) - Ordinal Regression analysis, Discussion and Adequacy of findings
3) W.A.D.Fernando (s16979) - Literature Review and Ordinal Regression analysis
4) J.B.A.A.Y Dharmasena (s16938) - Introduction and Descriptive Data analysis
5) O.S. Jayathunga (s16807) - Descriptive Data analysis , Literature Review and Conclusion

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